

Class 34: Review

Math 240

Exam 3

Fall 2011

1. (12 pts) A is a 11×8 matrix with rank 6. Give the dimension of these subspaces.

a) $\text{Col}(A)$ b) $\text{Col}(A^T)$ c) $\text{Row}(A)$ d) $\text{Row}(A^T)$ e) $\text{Nul}(A)$ f) $\text{Nul}(A^T)$

2. (6 pts) Let A be a 11×8 matrix with rank 6 and let T be the linear transformation defined by $T(\mathbf{x}) = A\mathbf{x}$. Determine if each of the following statements is true or false.

a) If $A\mathbf{x} = \mathbf{b}$ has a solution then that solution is unique.

b) $A\mathbf{x} = \mathbf{b}$ has a solution for every choice of $\mathbf{b}$ in $\mathbf{R}^{12}$.

c) T is a one-to-one function.

3. (12pts) Let $A = \begin{bmatrix} -2 & 4 \\ 3 & -1 \end{bmatrix}$.

a) What is the characteristic equation for A?

b) What are the eigenvalues for A? Show your work.

4. (16 pts) Let $A = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 2 & 0 \\ 4 & 0 & 0 \end{bmatrix}$. The eigenvalues of A are 2 and -2.

Is A diagonalizable? If so find a matrix P such that $P^{-1}AP = D$.

5. (10 pts) Let A be an $n \times n$ matrix with eigenvalue λ and let $ES = \{x: Ax = \lambda x\}$. Prove that ES is a subspace of $\mathbf{R}^n$.

6. (10 pts) Let A and B be $n \times n$ matrices. If A is similar to B and B is diagonalizable, prove that A is diagonalizable.

8. (12 pts) The first three Laguerre polynomials are 1 , $1 - t$, and $2 - 4t + t^2$.

a) Show that these polynomials form a basis, $\mathbf{B}$, for the polynomial space $\mathbf{P}_2$.

b) Find the coordinate vector for $11 - 23t - 5t^2$ with respect to the standard basis $\{1, t, t^2\}$.

c) What is the coordinate vector for $11 - 23t - 5t^2$ with respect to the basis $\mathbf{B}$?

9. (10 pts) Let A be a 3×3 matrix with eigenvalues 1, -1, and 2 and corresponding

eigenvectors $v_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $v_2 = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$, and $v_3 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$. Write $e_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ as a linear combination

of v_1, v_2 , and v_3 and use the linear combination to compute Ae_1 to find the first column of A.

3. (10 pts) A is a 9×12 matrix with rank 7. Give the dimension of these subspaces.

- a) $\text{Col}(A)$ b) $\text{Col}(A^T)$ c) $(\text{Row}(A))^\perp$ d) $\text{Null}(A^T)$ e) $\text{Null}(A)$

4. (6 pts) If A is an invertible matrix with eigenvalue λ and corresponding eigenvector $\mathbf{x}$ prove that $\frac{1}{\lambda}$ is an eigenvalue for A^{-1} .

5. (10 pts) Consider the matrix $A = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 5 & -10 \\ 1 & 0 & 2 & 0 \\ 1 & 0 & 0 & 3 \end{bmatrix}$ with eigenvalues 1, 2, and 3.

a) (6 pts) Find a basis for the eigenspace corresponding to $\lambda_1=1$.

b) (4 pts) Is the matrix A diagonalizable or not? Give a reason for your answer.

6. (12 pts) Let $\mathbf{w}_1 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}$ and $\mathbf{w}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$ and let $W = \text{span}(\mathbf{w}_1, \mathbf{w}_2)$.

a) Find a vector $\mathbf{v}$ that is in W and is orthogonal to $\mathbf{w}_1$.

b) Find a basis for the orthogonal complement $W^\perp$

7. (8 pts) Show that the set of vectors $\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} \right\}$ form an orthogonal basis for

$\mathbb{R}^3$ and find the coordinate vector of $\begin{bmatrix} -1 \\ 2 \\ 5 \end{bmatrix}$ with respect to that basis.

8. (10 pts) Let $\mathbf{v} = \begin{bmatrix} -1 \\ 10 \\ 3 \end{bmatrix}$ and $W = \text{span} \left(\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right)$. Find the orthogonal decomposition of $\mathbf{v}$ with respect to W . $\mathbf{v} = \mathbf{w} + \mathbf{w}^\perp$

9. (10 pts) Recall that $\mathcal{P}_2$ is the set of all polynomials with real coefficients and degree at most 2. Let $W = \{a + bx + cx^2 : a+b+c = 0\}$. Prove that W is a subspace of $\mathcal{P}_2$.

10. (8pts) The set of polynomials $\{1+x, x+x^2, 1+x^2\}$ is a basis for $\mathcal{P}_2$. Find the coordinate vector for $1+2x+3x^2$ with respect to this basis.

11. (6 pts) Suppose A and B are $n \times n$ orthogonal matrices. Prove that the product AB is orthogonal.

